Validation Roadmap for the Prior-Informed Bootstrap Framework

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Companion document to the Framework Specification
For developer and collaborator review

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Abstract

This document specifies the validation strategy for the prior-informed bootstrap framework introduced in the companion specification. It defines the network corpus on which the framework will be tested, the missingness mechanisms used to degrade fully observed networks into incomplete observations, the comparison baselines, the evaluation criteria, and the success conditions that determine whether the framework is ready for publication and broader use. The validation design draws methodologically from Smith et al. (2022), adapting their Monte Carlo protocol for the framework's distinctive credible-interval output. The document is intended as a commitment device: the conditions and criteria specified here are the ones the framework will be evaluated against, settled before implementation begins.

1 Introduction and Scope

The framework specified in the companion document introduces several methodological innovations relative to prior bootstrap approaches for network credible intervals — explicit node-level missingness, a user-supplied prior on centrality-missingness correlation, structural-role binning via the E/I index, and rank-conditional reciprocity imputation for nominated non-respondents. Each of these innovations expands the framework's applicability beyond what existing methods can handle, but also introduces additional design choices and assumptions that must be validated empirically before the framework can be recommended for general use.

This document specifies how that validation will be conducted. It defines:

- A network corpus combining empirical and synthetic networks spanning size, directedness, and centralization;
- Procedures for degrading fully observed networks into incomplete observations under controlled missingness mechanisms;
- Comparison baselines selected to represent the realistic alternatives a practitioner would consider;
- Evaluation criteria that capture both the framework's Bayesian credible-interval output and its compatibility with point-estimate comparisons established in prior work;
- Concrete success conditions against which the framework's performance will be judged.

The validation explicitly does not aim to dominate every existing imputation method in every setting. Prior work (Smith et al., 2022) has established that no single imputation approach is uniformly best across networks, measures, and missingness mechanisms. The framework's value proposition is providing well-calibrated credible intervals under non-random missingness, a capability that prior simple-imputation methods do not offer at all. The validation is therefore structured to demonstrate three claims: (1) the framework provides credible intervals with reasonable coverage and useful width across a range of realistic conditions; (2) the framework's point estimates are competitive with simple imputation methods on the metrics those methods were designed to optimize; and (3) the framework's calibration degrades gracefully when the user's supplied prior departs from the true missingness mechanism.

2 Validation Philosophy

The validation rests on three principles.

Treat empirical networks as ground truth. For each empirical network in the corpus, the network as collected is treated as the "true" network for evaluation purposes. We do not attempt to correct for any measurement error or missingness present in the original data collection; rather, we induce additional, controlled missingness on top of the original data and measure how well the framework recovers the original. This is the standard Monte Carlo paradigm in the network sampling literature (see Smith et al., 2022) and produces results that are interpretable as "how well does the framework recover the network the analyst would have if they had complete data?"

Use synthetic networks for controlled conditions. The empirical networks span specific structural cells but cannot be modified to test how performance varies along a single structural axis. Synthetic networks let us hold most properties constant and vary one at a time — size, centralization, density, community structure. The synthetic networks are not meant to replace the empirical ones; they complement them by providing controlled stress tests.

Match the methodological precedent where possible. Where Smith et al. (2022) established procedures for inducing missingness and evaluating bias, this validation follows those procedures. This produces results directly comparable to that prior work and avoids re-litigating methodological choices that have already been validated in the literature. Where the framework's distinctive features require new evaluation procedures (e.g., for credible-interval coverage), this document specifies those new procedures explicitly.

3 Network Corpus

The corpus comprises sixteen networks spanning the four cells of the weighted $\times$ directed design: five empirical, four synthetic at base specification, two thresholded unweighted variants of empirical networks, and four unweighted synthetic networks. The empirical networks cover real-world structural complexity at sizes ranging from 70 to 6,486 nodes. The synthetic networks fill structural cells the empirical corpus does not cover and provide controlled-condition baselines. The unweighted variants ensure the framework's distinctive ability to handle unweighted networks — absent in Bellutta (2026)'s Chapter 4 framework — is exercised across both directed and undirected settings. The Marvel co-appearance network ($N = 6{,}486$) was added as a large undirected representative, extending the corpus's size range by nearly a factor of five beyond the next-largest network and providing a large-and-clustered undirected counterpart to the large-and-sparse directed Balikatan case.

3.1 Empirical Networks

Moreno High School (directed, $N = 70$). Friendship nominations among boys in an Illinois high school, aggregated across fall 1957 and spring 1958 surveys. Edge weights range 1–2 indicating whether each boy was nominated in one or both surveys. Density 7.6%, mean degree 10.5, reciprocity 0.50, clustering coefficient 0.40. The network is small, dense, and highly reciprocated, with moderate clustering. It represents the canonical case where simple probabilistic imputation (using reciprocity as the imputation signal) is expected to work well, and serves as a natural test case for the framework's rank-conditional reciprocity matrix R.

Toledo Crime Networks (undirected, $N = 77$). Multiplex criminal cooperation network covering seven crime types: drug trafficking, theft, weapon carrying, homicide, kidnapping, arms trafficking, and ideological falsehood (Toledo et al., 2023). For this validation, the seven layers are collapsed to a single weighted summary network via sum-across-layers, following the procedure used in Bellutta (2026) for evaluating single-layer methods on multiplex data. The collapsed network has 223 weighted edges and density approximately 7–8%. Provides a small undirected empirical case with multiplex structural origin.

Scotland Interlock (undirected, $N = 108$). Bipartite director-company affiliation network for Scottish joint stock companies in 1904–1905, derived from Scott and Hughes (1980). For this validation, the bipartite network is projected to a one-mode company network: two companies are connected if they share one or more directors, weighted by the number of shared directors. The projection produces 108 nodes with company-level attributes (industry category, total capital). Provides an empirical case with known community structure (industry categories) that can serve as ground truth for Louvain community detection validation.

Balikatan 2022 (directed, $N = 1{,}347$). Twitter communication network from the Balikatan 2022 US-Philippines military exercise. Edges aggregate mentions, retweets, replies, and quotes between accounts; edge weights are integer interaction counts. Density approximately 0.17% (sparse), transitivity 0.23, modularity (Leiden) 0.56. Provides the corpus's only large, sparse empirical case with strong community structure. Most relevant for testing the framework's scalability and the value of E/I binning in a setting where community structure is strong but the network is too large for model-based imputation.

Marvel Universe (undirected, $N = 6{,}486$). Character co-appearance network derived from Marvel comic book data: two characters are connected if they appear together in the same comic, weighted by the number of shared appearances. With 77,227 edges, mean degree 11.9, and transitivity 0.79, it is the corpus's largest and most heavily clustered network, and its weight distribution is the most dispersed (weights range 2–744, Gini 0.54). The largest connected component spans 70% of nodes. Marvel is the corpus's large-undirected representative: it stress-tests the framework's scalability on a network nearly five times the size of Balikatan, in the undirected-and-clustered regime that complements Balikatan's directed-and-sparse one. Its heavy-tailed degree and weight distributions also make it the most demanding case for the weighted path-based measures and the layered triad census.

3.2 Synthetic Networks

The synthetic networks are designed in pairs, with each pair sharing an underlying structure but generated in both directed and undirected variants. This produces a clean experimental contrast between directionality holding other properties constant. Synthetic networks are generated with controlled parameters but realistic structural properties; specific parameter values may be refined after preliminary analysis of the empirical corpus (see Section 3.5).

Synthetic 1 (low centralization, $N = 600$). Stochastic block model with 4–6 communities of approximately equal size and moderate inter-community connectivity. Target density 5–7%, target clustering coefficient 0.30–0.40, target in-degree centralization 0.05–0.10. Edge weights drawn from Poisson($\lambda = 2$). Generated as a directed network; the undirected variant is produced by symmetrizing (union of forward and reverse edges, with weights summed). Provides a controlled large network where community structure is known by construction (allowing direct validation of Louvain recovery) and centralization is low (testing the framework where bin-exchangeability should hold most cleanly).

Synthetic 2 (high centralization, $N = 250$). Directed preferential attachment with attachment proportional to in-degree only, producing extreme in-degree concentration with uniform out-degree. Target density 2–3%, target in-degree centralization 0.25–0.30, target Bonacich centralization 0.35–0.45, target reciprocity 0.10–0.20 (low). Edge weights drawn from Poisson($\lambda = 2$). Undirected variant produced by symmetrization. Provides the corpus's controlled high-centralization case — the structural opposite of Moreno — and a stress test for bin-exchangeability when degree distribution is highly skewed.

3.3 Unweighted Variants

The framework's ability to handle unweighted networks is one of its distinctive capabilities relative to Bellutta (2026)'s Chapter 4 framework. To validate this capability across the four cells of the weighted $\times$ directed design, unweighted variants of four networks are added to the corpus, providing at least three networks in each cell.

Moreno (unweighted directed). Produced by thresholding the weighted Moreno network: any directed pair with weight ≥ 1 becomes an unweighted edge. The resulting network has the same edge presence pattern as the weighted version but loses the distinction between single and double nominations.

Toledo (unweighted undirected). Produced by thresholding the collapsed weighted summary: any unordered pair with at least one observed cooperation across the seven crime layers becomes an unweighted edge. The unweighted variant arguably aligns more closely with the underlying data semantics, since the original layer files record presence/absence of cooperation rather than intensity.

Synthetic 1 (unweighted directed and undirected). Generated using the same SBM specification as the weighted variants but with binary edges (Bernoulli draws on the block probability matrix, no Poisson weight assignment). The unweighted directed and undirected variants are generated as a pair through the same symmetrization procedure as the weighted versions. Provides controlled unweighted networks at scale ($N = 600$).

Synthetic 2 (unweighted directed and undirected). Generated using the same preferential attachment specification as the weighted variants but with binary edges. Provides controlled high-centralization unweighted networks at medium scale ($N = 250$).

A note on thresholded vs. natively-unweighted networks. Thresholding a weighted network produces a structurally different network than one that was unweighted by construction: edge density tends to be higher (any non-zero weight becomes an edge), reciprocity is computed on edge presence rather than weight, and degree distributions reflect edge counts rather than weight sums. The validation interprets unweighted variants as their own networks rather than as representations of the weighted versions, and reports results separately rather than treating them as paired observations of the same underlying structure.

3.4 Corpus Summary

The corpus brings together five empirical networks and two parameterized synthetic families spanning four substantive structural conditions. **Moreno High School** is a friendship network among 70 adolescents collected via standard sociometric instrument, representing the small-but-densely-connected case characteristic of school-based peer studies and providing reciprocity structure with which to test directed-graph measures. **Balikatan 2022** is a counter-insurgency interaction network derived from joint US–Philippines military exercise records ($N = 1{,}347$, $|E| = 3{,}146$), representing the large-and-sparse case that arises in operational social-network analysis where ties are recorded across many actors but at low density per actor. **Marvel Universe** is a character co-appearance network ($N = 6{,}486$, $|E| = 77{,}227$), representing the large-and-clustered undirected case: it is the corpus's largest network by a wide margin, with high transitivity (0.79) and a heavy-tailed weight distribution (Gini 0.54), and it complements Balikatan by occupying the opposite topology corner (dense and clustered rather than sparse). **Toledo Crime** is a co-offending network among 77 individuals appearing together in police records, representing the small-and-undirected case typical of administrative-data social networks where tie direction is unavailable or ill-defined. **Scotland Interlock** is a corporate-board interlock projection on 108 directors with ground-truth community structure available from the original bipartite affiliation data, providing a network where the validation framework's recovered partition can be compared against a known true partition. Each empirical network is included with both its native edge weights and a binary version where weights are stripped; the validation interprets these as separate networks rather than as paired observations. The two synthetic families — a stochastic block model (Synthetic 1) and a preferential attachment generator (Synthetic 2), each emitted in directed and undirected variants — anchor controlled low- and high-centralization endpoints, bracketing the empirical centralization range. Together the corpus covers the four cells of the weighted$\times$directed design and the full range of low-to-high degree centralization, plus extreme cases at the large-sparse (Balikatan), large-clustered (Marvel), and small-tree-like (Toledo, bicomponent fraction 0.27) corners of the topology space.

The sixteen GraphML files reside in the project's processed-data directory and have been validated by round-trip serialization tests (write to GraphML, load from GraphML, compare against in-memory representation): all node and edge attributes, weights, and graph-level metadata round-trip cleanly. The synthetic networks additionally satisfy their design specifications: Synthetic 1 has Leiden modularity 0.50 with six recovered communities at $\gamma = 1.0$, matching the six planted blocks; Synthetic 2 has maximum degree 43 and a top-five degree distribution of (43, 38, 35, 18, 13), confirming the hub-spoke structure expected of preferential attachment.

3.5 Phase 0: Preliminary Corpus Characterization

As a preliminary step before the main validation runs, Phase 0 computes descriptive statistics for all networks in the corpus using the measures from Smith et al. (2022): in/out/symmetric degree centrality, closeness, betweenness, Bonacich power, the corresponding centralizations, transitivity, mean inverse distance, component and bicomponent structure, and the ranked-cluster tau statistic. Phase 0 produces an empirical analog of Smith et al. (2022)'s Table 1 for the validation corpus, extended with weighted and triad-census characterizations that the framework's distinctive capabilities require.

Phase 0 reporting is organized into the opening corpus inventory (Table 1) plus eight descriptive tables. The first four split the centrality-and-topology battery across the 2×2 of directedness $\times$ weighting (Tables 2–5); the next two report the static (unweighted) triad census split by directedness (Tables 6–7); the last two report the weighted layered triad census, again split by directedness (Tables 8–9). This structure supersedes the single binarized table of earlier drafts, which relied on a binarization-invariance footnote to fold weighted and unweighted variants into one column. Separating binary from weighted explicitly is necessary now

Network	Source	N	Direction	Weights	$\lvert E\rvert$	Gini	Role
Moreno High School	Empirical	70	Directed	Weighted	366	0.24	Small directed dense
Moreno (unweighted)	Threshold	70	Directed	Unweighted	366	0.24	Unweighted directed
Toledo Crime (collapsed)	Empirical	77	Undirected	Weighted	106	0.46	Small undirected real
Toledo (unweighted)	Threshold	77	Undirected	Unweighted	106	0.46	Unweighted undirected
Scotland Interlock (proj.)	Empirical	108	Undirected	Weighted	276	0.48	Comm. truth available
Balikatan 2022	Empirical	1,347	Directed	Weighted	3,146	0.70	Large directed sparse
Marvel Universe	Empirical	6,486	Undirected	Weighted	77,227	0.54	Large undirected clustered
Marvel (unweighted)	Threshold	6,486	Undirected	Unweighted	77,227	0.54	Unweighted undir. at scale
Synthetic 1 (directed)	SBM	600	Directed	Weighted	8,277	0.10	Large directed controlled
Synthetic 1 (undirected)	SBM sym.	600	Undirected	Weighted	4,138	0.14	Large undirected controlled
Synthetic 1 (dir., unwt.)	SBM bin.	600	Directed	Unweighted	8,253	0.10	Unwt. dir. at scale
Synthetic 1 (undir., unwt.)	SBM bin. sym.	600	Undirected	Unweighted	4,228	0.14	Unwt. undir. at scale
Synthetic 2 (directed)	PA	250	Directed	Weighted	497	0.38	Mid directed extreme
Synthetic 2 (undirected)	PA sym.	250	Undirected	Weighted	497	0.38	Mid undirected extreme
Synthetic 2 (dir., unwt.)	PA bin.	250	Directed	Unweighted	497	0.37	Unwt. high-cent. dir.
Synthetic 2 (undir., unwt.)	PA bin. sym.	250	Undirected	Unweighted	497	0.37	Unwt. high-cent. undir.

Table 1: Validation network corpus, 16 networks across the four weighted $\times$ directed cells. Edge counts ($|E|$) and Gini coefficients are computed from the assembled GraphML files. The Gini column reports the coefficient of the unweighted-degree distribution for the synthetic and unweighted-empirical rows, and of the edge-weight distribution for the weighted-empirical rows where noted; < 0.20 indicates low centralization (uniform degree), > 0.35 indicates high centralization (hub-skewed degree). The empirical corpus spans the full range from uniform to highly hub-skewed and adds Marvel at the large-clustered extreme. Synthetic 1 anchors the low-centralization endpoint (Gini 0.10–0.14); Synthetic 2 anchors the high-centralization endpoint (Gini 0.37–0.38). Together the corpus tests the validation framework across a broad swath of the structural-coverage space.

that the framework computes genuinely weighted path-based measures (closeness, betweenness, and mean inverse distance under tie-strength edge semantics), whose values differ from their binarized counterparts and could not be represented in a binarization-invariant table. The triad census is split by directedness because the directed and undirected cases have different sets of non-zero Davis–Leinhardt classes (sixteen versus four) and dramatically different τ-grid scales, which would render uninformatively if pooled.

Within each of the four measure tables, the empirical networks appear in the leftmost columns and the synthetic networks in the rightmost columns, separated by a vertical rule. This grouping makes the design argument legible at a glance: the empirical networks span a wide range of structural conditions, and the synthetic networks anchor controlled low- and high-centralization endpoints of that range. Reading horizontally across an empirical-then-synthetic row shows whether the empirical variation lies within, outside, or astride the synthetic controls.

Binary versus weighted reporting conventions. The two binary tables (2, 3) compute every measure on the binarized adjacency, matching the convention of Smith et al. (2022)'s Table 1 and ensuring cross-network comparability; the weighted and unweighted variants of a network are binarization-invariant and so each network appears once. The two weighted tables (4, 5) report measures that depend on edge weights and therefore include only the genuinely weighted networks: the unweighted variants would collapse to their binary values (every weight equal to one), so they are omitted rather than shown as redundant or degenerate rows. Two weight-handling conventions are worth making explicit, because they differ across the table families. First, the weight-distribution summaries (min, max, median, IQR, Gini) are computed on the *aggregated directed edge weights* for directed networks and on the *nonzero weights of the symmetrized adjacency's upper triangle* for undirected networks — the latter matching the graph that the undirected weighted path-based measures actually traverse, where reciprocal arcs are combined by $\max(A, A^\top)$. Second, the weighted path-based measures symmetrize undirected graphs by $\max(A, A^\top)$, whereas the weighted triad census symmetrizes by sum $(W + W^\top)$ before thresholding, following *s*-core semantics; the two table families therefore treat undirected edge weights differently by design, and values should not be expected to align across them.

	Empirical			*Synthetic*	
Measure	Marvel	Scot	Toledo	SBM-u	PA-u
Centrality, mean (SD)					
Degree	11.91 (31.71)	2.56 (2.31)	1.38 (1.70)	7.05 (1.75)	1.99 (2.28)
Closeness	0.19 (0.12)	0.26 (0.14)	0.34 (0.07)	0.38 (0.02)	0.32 (0.05)
Betweenness	0.00 (0.00)	0.01 (0.02)	0.03 (0.10)	0.00 (0.00)	0.01 (0.03)
Bonacich Power	0.03 (0.07)	0.19 (0.19)	0.14 (0.14)	0.51 (0.13)	0.10 (0.11)
Centralization (SD of node scores)					
Degree	31.71	2.31	1.70	1.75	2.28
Closeness	0.13	0.14	0.07	0.02	0.05
Betweenness	0.00	0.02	0.10	0.00	0.03
Bonacich Power	0.07	0.19	0.14	0.13	0.11
Topology					
Component (prop.)	0.70	0.80	1.00	1.00	1.00
Bicomponent (prop.)	0.65	0.73	0.27	1.00	1.00
Mean Inv. Distance	0.05	0.06	0.05	0.13	0.02
Transitivity	0.79	0.46	0.42	0.06	0.10

Table 2: Phase 0 Table 1 — undirected, binary. Centrality and topology measures computed on the binarized adjacency for the undirected networks, empirical (left) and synthetic (right). Each column is a single network; weighted and unweighted variants are binarization-invariant and so appear once. Centrality cells show mean (SD) of node-level scores; centralization is the network-level standard deviation of those scores, following Smith et al. (2022). Component and bicomponent sizes are reported as proportions of the node set in the largest (bi)component. Marvel anchors the large-clustered extreme (transitivity 0.79, $N = 6{,}486$); Synthetic 1 (SBM-u) and Synthetic 2 (PA-u) anchor the low- and high-centralization synthetic endpoints.

	Empirical		*Synthetic*	
Measure	Moreno	Bali	SBM-d	PA-d
Centrality, mean (SD)				
In-Degree	5.23 (3.64)	2.34 (2.70)	13.76 (3.52)	1.99 (4.65)
Out-Degree	5.23 (2.04)	2.34 (18.55)	13.76 (3.36)	1.99 (0.14)
Symmetric Degree	10.46 (4.56)	4.64 (19.11)	27.51 (4.98)	3.98 (4.55)
Closeness	0.45 (0.06)	0.23 (0.12)	0.45 (0.02)	0.32 (0.05)
Betweenness	0.04 (0.05)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Bonacich Power	0.39 (0.19)	0.02 (0.04)	0.58 (0.11)	0.10 (0.11)
Centralization (SD of node scores)				
In-Degree	3.64	2.70	3.52	4.65
Out-Degree	2.04	18.55	3.36	0.14
Symmetric Degree	4.56	19.11	4.98	4.55
Closeness	0.06	0.12	0.02	0.05
Betweenness	0.05	0.00	0.00	0.00
Bonacich Power	0.19	0.04	0.11	0.11
Topology				
Component (prop.)	1.00	0.81	1.00	1.00
Bicomponent (prop.)	1.00	0.49	1.00	1.00
Mean Inv. Distance	0.45	0.23	0.45	0.32
Transitivity	0.40	0.23	0.05	0.06
Tau$_{RC}$	8.73	2535.93	−6.82	63.03

Table 3: Phase 0 Table 2 — directed, binary. Measures computed on the binarized adjacency for the directed networks, empirical (left) and synthetic (right). Centrality cells show mean (SD); centralization is the network-level SD of node scores. Component and bicomponent sizes are proportions of the node set. Mean inverse distance is the mean reciprocal geodesic distance over ordered pairs. Tau$_{RC}$ is the standardized deviation of the observed ranked-cluster-weighted triad count from its U|MAN null expectation, by Monte Carlo with 500 samples; its magnitude scales with $\sqrt{N}$ for very-sparse networks, explaining the large Balikatan value. Tau$_{RC}$ is defined only for directed graphs (it requires the mutual–asymmetric dyad distinction) and so appears in this table but not Table 2.

	Empirical			*Synthetic*	
Measure	Marvel	Scot	Toledo	SBM-u	PA-u
Weight distribution					
Minimum	2.00	1.00	1.00	1.00	1.00
Maximum	744.00	8.00	5.00	9.00	6.00
Median	3.00	1.00	2.00	3.00	2.00
IQR	3.00	0.00	2.00	2.00	2.00
Gini	0.54	0.24	0.26	0.26	0.26
Centrality, mean (SD)					
Strength	73.81 (333.28)	3.45 (3.63)	2.90 (4.14)	20.65 (5.88)	3.92 (4.24)
Wtd. Closeness	1.28 (1.18)	0.30 (0.17)	0.67 (0.24)	1.30 (0.09)	0.60 (0.12)
Wtd. Betweenness	0.00 (0.00)	0.01 (0.02)	0.03 (0.10)	0.00 (0.00)	0.01 (0.03)
Bonacich Power	0.03 (0.07)	0.19 (0.19)	0.14 (0.14)	0.56 (0.14)	0.11 (0.11)
Centralization (SD of node scores)					
Strength	333.28	3.63	4.14	5.88	4.24
Wtd. Closeness	1.18	0.17	0.24	0.09	0.12
Wtd. Betweenness	0.00	0.03	0.10	0.00	0.04
Bonacich Power	0.07	0.19	0.14	0.14	0.11
Topology					
Wtd. Mean Inv. Distance	0.28	0.07	0.09	0.40	0.04

Table 4: Phase 0 Table 3 — undirected, weighted. Weighted measures for the genuinely weighted undirected networks; unweighted variants are omitted because their weighted measures collapse to the binary values of Table 2. Weighted closeness, betweenness, and mean inverse distance use tie-strength edge semantics (Dijkstra on $1/w$, so stronger ties are shorter). Strength is weight-summed degree. The weight-distribution block (min, max, median, IQR, Gini) is computed on the nonzero weights of the symmetrized adjacency's upper triangle ($\max(A, A^\top)$), the graph the undirected weighted measures traverse. Marvel's dispersed weights (max 744, Gini 0.54) drive its high mean strength and weighted closeness relative to the binary case.

Measure	*Empirical* Moreno	Bali	*Synthetic* SBM-d	PA-d
Weight distribution				
Minimum	1.00	1.00	1.00	1.00
Maximum	2.00	31.00	10.00	6.00
Median	1.00	1.00	3.00	2.00
IQR	1.00	0.00	2.00	2.00
Gini	0.17	0.27	0.26	0.26
Centrality, mean (SD)				
In-Strength	7.23 (5.38)	3.35 (6.75)	41.53 (11.78)	3.92 (8.60)
Out-Strength	7.23 (2.65)	3.35 (30.26)	41.53 (12.04)	3.92 (1.43)
Total Strength	14.46 (6.93)	6.62 (32.79)	83.07 (17.00)	7.83 (8.48)
Wtd. Closeness	0.59 (0.08)	0.35 (0.20)	1.69 (0.09)	0.60 (0.12)
Wtd. Betweenness	0.04 (0.05)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Bonacich Power	0.39 (0.19)	0.02 (0.04)	0.60 (0.11)	0.11 (0.11)
Centralization (SD of node scores)				
In-Strength	5.38	6.75	11.78	8.60
Out-Strength	2.65	30.26	12.04	1.43
Total Strength	6.93	32.80	17.00	8.48
Wtd. Closeness	0.08	0.20	0.09	0.12
Wtd. Betweenness	0.05	0.00	0.00	0.00
Bonacich Power	0.19	0.04	0.11	0.11
Topology				
Wtd. Mean Inv. Distance	0.59	0.35	1.69	0.60

Table 5: Phase 0 Table 4 — directed, weighted. Weighted measures for the genuinely weighted directed networks; unweighted variants are omitted (their weighted measures collapse to Table 3). Weighted path-based measures use tie-strength semantics (Dijkstra on $1/w$). In-, out-, and total strength are the weight-summed in-, out-, and total degree. The weight-distribution block is computed on the aggregated directed edge weights. Moreno's narrow weight range (1–2, recording single vs. double nominations) yields the lowest weight Gini; Balikatan's interaction counts are the most dispersed in the out-direction (out-strength centralization 30.26), reflecting a few very high-volume broadcasting accounts.

	Empirical		*Synthetic*	
DL Class	Moreno	Bali	SBM-d	PA-d
003	37,827	4.03×10^8	31,233,805	2,453,719
012	9,822	3,602,267	4,204,617	115,292
102	5,175	38,837	158,702	0
021D	170	225,413	47,815	196
021U	311	3,865	49,106	2,886
021C	323	14,258	101,315	849
111D	430	127	7,329	0
111U	214	3,334	6,997	0
030T	82	3,400	6,648	58
030C	5	2	2,369	0
201	67	15	227	0
120D	89	65	277	0
120U	41	669	247	0
120C	36	3	684	0
210	92	9	61	0
300	56	5	1	0

Table 6: Phase 0 Table 5a — static triad census (directed networks), Davis–Leinhardt class counts. A single Batagelj–Mrvar census on the binarized graph; each cell is the raw count of triples in that class. All sixteen DL classes appear; counts span several orders of magnitude and are reported with thousand separators (scientific notation for cells $\geq 10^8$). The closed-triangle count (class 300) matches the unweighted triangle count $T_{\max}$ used by the layered census in Table 8.

	Empirical			*Synthetic*	
DL Class	Marvel	Scot	Toledo	SBM-u	PA-u
003	4.50×10^{10}	176,906	65,795	33,349,744	2,453,752
102	4.74×10^8	25,513	6,789	2,413,690	115,292
201	11,718,673	1,468	537	55,644	3,904
300	1,028,453	269	29	1,122	52

Table 7: Phase 0 Table 5b — static triad census (undirected networks), Davis–Leinhardt class counts. Only the four undirected isomorphism classes $\{003, 102, 201, 300\}$ are non-zero on undirected graphs; the twelve asymmetric DL classes are zero by construction and omitted. Counts are computed by the dyad-driven Batagelj–Mrvar kernel applied to the symmetrized adjacency, validated against an external Pajek reference on Balikatan and against the reference triple-loop kernel on Scotland. Marvel's $300 = 1{,}028{,}453$ closed triangles match the unweighted triangle count $T_{\max}$ used by the layered census in Table 9.

	Empirical		*Synthetic*	
Quantity	Moreno	Bali	SBM-d	PA-d
Grid metadata				
$T_{\max}$	307	3,015	7,001	42
L	10	8	8	16
$\tau_{\min}$	1.00	1.26	2.15	1.00
$\tau_{\max}$	2.00	4.97	4.64	5.00
Method	empirical	analytic	analytic	empirical
Wall-clock (s)	0.07	0.24	0.06	0.02
Per-class AUMC and peak τ				
AUMC_{003}	0.259	0.596	0.315	0.687
peak τ_{003}	1.08	4.08	4.16	5.00
AUMC_{012}	0.0271	5.63×10^{-4}	0.0179	0.0117
peak τ_{012}	1.00	1.26	2.15	1.00
AUMC_{102}	0.0124	1.05×10^{-5}	3.22×10^{-4}	0
peak τ_{102}	1.00	1.26	2.15	1.00
AUMC_{021D}	9.35×10^{-5}	9.03×10^{-6}	1.05×10^{-4}	1.16×10^{-5}
peak τ_{021D}	1.00	1.26	2.15	1.00
AUMC_{021U}	3.70×10^{-4}	3.71×10^{-7}	1.06×10^{-4}	1.43×10^{-4}
peak τ_{021U}	1.00	1.26	2.15	1.00
AUMC_{021C}	3.84×10^{-4}	1.44×10^{-6}	2.23×10^{-4}	4.50×10^{-5}
peak τ_{021C}	1.00	1.26	2.15	1.00
AUMC_{111D}	5.05×10^{-4}	4.37×10^{-8}	8.52×10^{-6}	0
peak τ_{111D}	1.00	1.26	2.15	1.00
AUMC_{111U}	1.69×10^{-4}	1.93×10^{-7}	8.46×10^{-6}	0
peak τ_{111U}	1.00	1.26	2.15	1.00
AUMC_{030T}	6.66×10^{-5}	2.84×10^{-7}	7.72×10^{-6}	1.58×10^{-6}
peak τ_{030T}	1.00	1.26	2.15	1.00
AUMC_{030C}	1.53×10^{-6}	0	2.79×10^{-6}	0
peak τ_{030C}	1.00	1.26	2.15	1.00
AUMC_{201}	7.24×10^{-5}	2.94×10^{-9}	1.75×10^{-7}	0
peak τ_{201}	1.00	1.26	2.15	1.00
AUMC_{120D}	1.52×10^{-4}	2.62×10^{-9}	1.84×10^{-7}	0
peak τ_{120D}	1.00	1.26	2.15	1.00
AUMC_{120U}	5.93×10^{-5}	2.11×10^{-8}	1.26×10^{-7}	0
peak τ_{120U}	1.00	1.26	2.15	1.00
AUMC_{120C}	2.66×10^{-5}	1.47×10^{-9}	5.20×10^{-7}	0
peak τ_{120C}	1.00	1.26	2.15	1.00
AUMC_{210}	1.16×10^{-4}	9.44×10^{-10}	3.26×10^{-8}	0
peak τ_{210}	1.00	1.26	2.15	1.00
AUMC_{300}	4.83×10^{-5}	0	0	0
peak τ_{300}	1.00	1.26	2.15	1.00

Table 8: Phase 0 Table 6a — weighted layered triad census (directed networks). For each network the layered census thresholds the weighted adjacency across a log-spaced grid of L thresholds $\tau \in [\tau_{\min}, \tau_{\max}]$ chosen by `recommend_L`; per-class AUMC values are areas under the density curves on log-τ, and peak τ is the threshold at which each class's density peaks. AUMC values span several orders of magnitude (the empty-triad class 003 dominates while the dense motifs 201 and 300 are rare) and are reported in scientific notation; peak τ is reported in the native τ units. The method dispatch (analytic vs. empirical) is set by `recommend_L` based on $T_{\max}$ and the weight-distribution span. Unweighted variants of these networks are omitted because their weighted measures collapse to the static counts of Table 6.

Quantity	Empirical			Synthetic	
	Marvel	Scot	Toledo	SBM-u	PA-u
Grid metadata					
$T_{\max}$	1,028,453	269	29	1,024	42
L	14	16	10	16	16
$\tau_{\min}$	4.40	1.00	1.00	1.00	1.00
$\tau_{\max}$	227.75	8.00	4.00	7.00	5.00
Method	analytic	empirical	empirical	empirical	empirical
Wall-clock (s)	1.59	0.01	0.00	0.14	0.02
Per-class AUMC and peak τ					
AUMC_{003}	1.71	0.889	0.572	0.816	0.687
peak τ_{003}	227.75	5.28	3.43	6.15	4.03
AUMC_{102}	1.15×10^{-3}	0.0135	0.028	0.0286	0.0117
peak τ_{102}	4.40	1.00	1.00	1.00	1.00
AUMC_{201}	9.19×10^{-6}	2.81×10^{-4}	1.46×10^{-3}	4.71×10^{-4}	1.99×10^{-4}
peak τ_{201}	4.40	1.00	1.00	1.00	1.00
AUMC_{300}	1.31×10^{-6}	6.71×10^{-5}	1.19×10^{-4}	7.00×10^{-6}	1.58×10^{-6}
peak τ_{300}	4.40	1.00	1.00	1.00	1.00

Table 9: Phase 0 Table 6b — weighted layered triad census (undirected networks). Only the four undirected isomorphism classes $\{003, 102, 201, 300\}$ are non-zero on undirected graphs; the twelve asymmetric DL classes are zero across the τ grid by construction and omitted. Weights are symmetrized by sum $(W + W^\top)$ before thresholding, following s-core semantics — a different convention from the $\max(A, A^\top)$ symmetrization used by the weighted path-based measures in Tables 4–5. AUMC values are reported in scientific notation, peak τ in native units. Marvel's $T_{\max} = 1{,}028{,}453$ and analytic τ grid span the widest range in the corpus.

4 Degeneration Strategy

For each network in the corpus, the validation proceeds by treating the network as the true network T and generating degraded observed networks G_{obs} by removing nodes under a controlled missingness mechanism. The framework is then run on G_{obs} and its output compared against the known true measures computed on T. This section specifies the degeneration design; it is a design specification for the Phase 1 sampler, which has not yet been implemented.

4.1 A Two-Factor Degeneration Design

The degeneration design is fundamentally *two-factor*, and the distinction is central to the framework's value proposition. The two factors are not a nuisance grid to be marginalized over: they *are* the priors against which the framework's calibration is measured, and the sampler must realize both simultaneously in each degraded network.

Factor 1 — missingness rate. The quantity of nodes removed, expressed as a proportion of the node set. This sets *how many* nodes are dropped. For the main validation, six rates are used: 5%, 10%, 15%, 25%, 40%, and 50% of nodes. This covers the realistic range an analyst is likely to face and matches a subset of the rates explored in Smith et al. (2022). Lower rates (1–5%) are excluded because between-method differences are too small to detect reliably; higher rates (60–70%) are excluded as extreme conditions where any method's reliability is suspect.

Factor 2 — missingness–centrality correlation. The selectivity of removal: the correlation ρ between a node's centrality and its probability of being removed. This sets *which* nodes are more likely to drop — high-centrality nodes over-represented among the removed (positive ρ), under-represented (negative ρ), or removed at random ($\rho = 0$, the MCAR baseline). The induced correlation is set at five levels: -0.75, -0.25, 0.00, $+0.25$, $+0.75$. The ± 0.75 and ± 0.25 values match the four levels used in Smith et al. (2022); the 0.00 level provides an MCAR baseline.

The validation is therefore a two-dimensional grid over (rate, ρ), with $6 \times 5 = 30$ cells per network. The framework's claim is that it produces well-calibrated credible intervals across that grid, and the grid is precisely the space of priors the user is asked to supply.

4.2 The Joint Sampler

Missingness is induced at the node level. The probability that node i is selected for removal is a convex combination of a centrality signal and uniform noise:

$$\text{prob}_i = b \cdot \tilde{c}_i + (1 - b) \cdot u_i,$$

where $\tilde{c}_i$ is the centrality vector after min–max normalization to $[0, 1]$, $u_i \sim \text{Uniform}(0, 1)$, and the mixing scalar $b \in [0, 1]$ controls selectivity. The centrality driver is the binarized *in-degree* for directed networks and the binarized *degree* for undirected networks — following Smith et al. (2022) for the directed case and extending it naturally to the undirected case. Negative target ρ is handled by sign-flipping the probability vector ($\text{prob}_i \leftarrow 1 - \text{prob}_i$) before sampling, which preferentially selects low-centrality nodes while keeping all weights in $[0, 1]$.

The bisection target: realized indicator correlation, not probability correlation. The framework departs from Smith et al. (2022) in what quantity ρ controls. The prior work defined ρ as the correlation between centrality and the *probability* of being a non-respondent — a deterministic, sampler-recipe quantity that the bisection on b can target directly. The framework instead defines ρ as the expected realized correlation between the *dropped-status indicator* and centrality:

$$\rho_{\text{target}} = \mathbb{E}\left[\text{cor}\left(1_{\text{dropped}}, c\right)\right],$$

where the expectation is over the weighted-without-replacement draw given b, target rate, and the probability vector above. The dropped-indicator quantity is what the downstream analysis actually conditions on: it is the empirically measurable correlation between which nodes ended up missing and their centrality, not the latent recipe used to draw them. The two quantities agree in expectation over many replicates for non-extreme parameter regimes but differ systematically near the achievable ceiling; matching the analyst's mental model requires bisecting against the realized-indicator quantity.

Monte Carlo bisection. The realized-indicator target is not analytically invertible in b, so the bisection is run against a Monte Carlo estimate. At each candidate b the sampler draws $M = 50$ inner weighted-without-replacement samples at the target rate, computes the empirical mean of $\text{cor}(1_{\text{dropped}}, c)$ across them, and bisects that mean against $|\rho_{\text{target}}|$ (the sign of ρ is handled separately via the probability-vector flip). The bisection tolerance is set to 0.02 to accommodate the finite-sample noise of the M=50 inner estimate (standard error roughly 0.015 on networks at typical centrality variance). Each bisection iteration uses deterministic per-iteration seeds derived from the master seed and iteration index, so the procedure is reproducible from a single master seed regardless of thread scheduling.

Achievable-ρ ceiling. The attainable expected realized correlation between 1_{dropped} and centrality is bounded above by the structure of the centrality vector. On highly skewed networks — the preferential-attachment synthetics, the constructed star fixture used in validation testing — a few hubs dominate centrality and the probability vector becomes *saturated* at $b = 1$: only a small number of entries have positive weight. Bisecting toward $\rho = +0.95$ on such fixtures is infeasible. The sampler detects this by probing $b = 1$ at the start of every bisection: if the resulting Monte Carlo estimate of the indicator correlation falls below the target by more than the tolerance, the sampler returns the ceiling estimate, sets `bisection_status = :ceiling_hit`, and records the achieved realized ρ as the actual condition rather than reporting the unattained nominal target.

Saturation handling at extreme b. When the probability vector has fewer than $k = \lfloor \text{rate} \cdot N \rfloor$ strictly positive entries — the regime where $b = 1$ on a centrality-saturated fixture concentrates all weight on one or two nodes — standard weighted-without-replacement sampling cannot proceed: there are fewer positive weights than draws required. The sampler falls back to a deterministic-positive-plus-uniform-fill draw: all positive-weight nodes are selected, and the remaining $k - n_+$ slots are filled by uniform draws from the zero-weight nodes. This produces a well-defined dropped set whose realized indicator correlation is below any target ρ near 1, consistent with the bisection's `:ceiling_hit` classification. Both the Monte Carlo bisection's inner samples and the production sampler implement this fallback; on real corpus networks (Marvel, Balikatan, the synthetics) the saturation path does not fire because real centrality distributions have enough variance to support the full k positive entries at $b = 1$.

Recording nominal, realized, and degeneracy. The sampler always records three things for each degraded network: the *nominal* (rate, ρ) requested, the *realized* (rate, ρ) actually achieved by the sampled dropped-node set, and a *degeneracy flag*. Realized values differ from nominal even when the bisection succeeds in expectation, because each finite draw has sampling noise — severe at small N and low rates (5% of 70 nodes is three to four nodes, a very noisy realized correlation). Recording both lets coverage be tabulated against either the nominal grid or the realized conditions, and it feeds the per-cell degeneracy rate used for grid trimming below.

Degenerate draws are kept and flagged, not discarded. At high rates with strong positive ρ — for example 50% missingness with $\rho = +0.75$ — most high-centrality nodes are removed, which can disconnect the graph, collapse the giant component, or empty the E/I bins. This is the realistic outcome of selective high-rate missingness, especially for small-world structures, and the validation keeps these draws rather than discarding or re-rolling them. Each draw is flagged for degeneracy against concrete criteria (giant-component fraction below a threshold, empty bins, undefined measures), and the per-cell degeneracy rate is recorded alongside the nominal and realized conditions. That degeneracy rate is a planning instrument: cells that degenerate at high frequency need few replicates, so the degeneracy profile is used to set per-network run bounds and trim the grid, surfacing the hard cases rather than hiding them.

Constant centrality fails explicitly. When the centrality vector has no variance — the 4-regular ring fixture used in validation testing, or any pathological case where binarized degree is constant — no value of b can induce a non-zero correlation. The Monte Carlo bisection's inner correlation estimate returns NaN on the first iteration, the bisection short-circuits to `bisection_status = :failed_other`, and the record is marked as unusable. The caller (the orchestrator, or any downstream analysis) treats `:failed_other` records as excluded from coverage computation. This explicit failure mode replaces the prior design's "ties attenuate selectivity" provision: rather than reporting a degraded-but-finite realized ρ on degenerate inputs, the sampler is loud about the structural impossibility.

Seed discipline. With five ρ levels, six rates, up to 100 replicates, and sixteen networks, the grid involves a large number of draws. Each (network, ρ, rate, replicate) tuple is independently reproducible from a single master seed via a deterministic sub-seed scheme, so any one cell can be regenerated without rerunning the grid.

Replication. The per-cell replication count is 100 for small and mid-sized networks, with a smaller count permitted for Marvel where the per-replicate cost is highest; the degeneracy profile (above) further trims replication in cells that degenerate at high frequency.

API and cached centrality. The sampler is structured as a small family of internal helpers behind a public API. The centrality vector for a network is computed once and cached, since binarized in-degree (directed) or binarized degree (undirected) is a property of the input network that does not change across (ρ, rate) targets. (For undirected networks, the centrality computation sums both row and column entries of the adjacency, because the package's sparse-matrix builder stores each undirected edge as a single directed entry rather than symmetrizing.) The internal routines are `_centrality_for_sampler`, which builds the cached centrality vector; `_centrality_correlation_for_b`, which evaluates the Monte Carlo estimate of $\mathbb{E}[\text{cor}(1_{\text{dropped}}, c)]$ for a given b via M inner samples; the bisection routine that solves for b given a target ρ to within tolerance 0.02 (the looser-than-typical tolerance accommodates the M=50 inner-estimate standard error); `_sample_missingness`, which produces the dropped-node set given centrality, target rate, b, sign, and a seed; and `_topological_degeneracy`, which computes the giant-component, node-count, and edge-count summaries on the degraded network. The public API exposes `generate_missingness_mask` (wrapping the internals and returning the complete replicate record), `apply_missingness` (constructing the degraded network under the full-node-removal mechanism), `apply_missingness_outgoing_only` (the SMM-style mechanism that removes only outgoing edges and keeps the non-respondent in the roster), and `build_degeneration_corpus` (orchestrating the full grid across networks, ρ values, rates, replicates, and mechanisms).

4.3 Dual Materialization Mechanisms

The validation evaluates the framework under two distinct missing-data mechanisms drawn from the literature, each corresponding to a different scenario the framework was designed to handle. Both mechanisms share the same sampler — the same centrality-correlated dropped-node selection — and differ only in how the degraded network is materialized from the dropped-node set. This separation of selection from materialization is what makes a doubled-mechanism grid computationally tractable: the expensive bisection runs once per cell, both mechanisms reuse the result.

Mechanism A: full node removal (`:full_removal`). The dropped nodes are removed from the network entirely, along with all of their incident edges (both incoming and outgoing). The remaining network has $N - k$ nodes where k is the dropped count, and only edges between surviving nodes. This is the mechanism implied by the framework's Stage 1 (synthetic node insertion for non-nominated non-respondents): non-respondents are entirely unobserved, including by other respondents, and the framework reconstructs them from scratch. It is also the mechanism that corresponds to the *listwise deletion* baseline in Smith et al. (2022), where the analyst proceeds with only the observed subset.

Mechanism B: SMM-style outgoing-edge removal (`:outgoing_only`). The dropped nodes *remain* in the node roster, but their outgoing edges are removed; their incoming edges (nominations from observed actors who reported on them) are preserved. The remaining network has the same node set as the original but reduced edges. This is the actor-non-response mechanism specified in Smith et al. (2022): the non-respondent's existence is known because other actors named them, and the analyst has the in-edges but not the out-edges. It is the mechanism the framework's Stage 0.5 (rank-conditional reciprocity imputation) is designed to address.

Why both, rather than picking one. The two mechanisms test different parts of the framework. Full node removal stress-tests Stage 1's reconstruction of nodes that the analyst has no information about; outgoing-edge-only removal stress-tests Stage 0.5's reconstruction of edges from partially-observed actors. A framework that handles both — and a validation that exercises both — is more informative than a single-mechanism evaluation. Mechanism A is the default and applies to all networks; Mechanism B is defined only for directed networks (undirected networks have no "outgoing" direction by definition) and is silently skipped on undirected fixtures in the orchestrator with a startup diagnostic identifying which networks are skipped.

Selection invariance. Both mechanisms consume the *same* dropped-node set from the same sampler call. This is enforced by the orchestrator's design: the sampler runs once per $(\text{network}, \rho, \text{rate}, \text{replicate})$ cell, the dropped-node set is cached, and both materializers (when applicable) consume that cached set. The implication is that the topological-degeneracy fields recorded with each replicate are mechanism-agnostic: the giant-component fraction, surviving node count, and surviving edge count are computed against the original adjacency with the dropped-node mask, not against either materialized network. Two output rows for the same cell (one per mechanism) therefore share their degeneracy values but differ on the `mechanism` field and, when measures run downstream, on the measure-degeneracy fields specific to each materialized network.

Grid implications. The full validation grid, with both mechanisms enabled and treating undirected networks as Mechanism-A-only, expands from the single-mechanism size of $16 \times 5 \times 6 \times 100 = 48{,}000$ records to a doubled-grid size of approximately 72,000 records (the directed networks contribute two records per cell, undirected networks contribute one). The marginal computational cost of the second mechanism is small: only the materializer and the per-row record assembly differ; the sampler call — where most of the per-cell time is spent — is shared.

4.4 Recording Schema and Storage

Each replicate emits a complete record sufficient to reconstruct the degraded network and to support both nominal-grid and realized-condition analyses downstream. The record is the dropped-node index set together with realized-condition scalars and degeneracy flags; the full degraded network itself is *not* stored.

Dropped-node sets, not degraded networks. The full grid involves up to 48,000 degraded networks (sixteen networks $\times$ five ρ $\times$ six rates $\times$ 100 replicates), and storing each as a GraphML file would be prohibitive at Marvel scale and wasteful at every scale. The dropped-node index set, by contrast, is a vector of integers, a few kilobytes per replicate at most. Combined with the original network it reconstructs the degraded network exactly when `apply_missingness` is called at measurement time. This is the standard lazy-materialization pattern: store the difference from the canonical, not the result.

Per-replicate record. Each replicate's record carries:

- `dropped_nodes`: the vector of removed node indices.
- `nominal`: the target (ρ, rate) for the cell.

- `realized_rate`: the actual fraction removed (deterministic given target rate and N).
- `realized_rho`: the realized correlation between the dropped-status indicator and centrality across the node set — the empirically measurable quantity the bisection was solving for, signed against the original centrality direction so that negative-target draws return negative realized values.
- `bisection_status`: a three-way symbol indicating how the bisection terminated:
  - `:converged` — the Monte Carlo bisection hit target ρ within tolerance;
  - `:ceiling_hit` — the bisection's $b = 1$ probe estimate was below target by more than the tolerance, indicating the centrality distribution cannot support the target;
  - `:failed_other` — structural failure (e.g., constant centrality, where the inner correlation estimate returns NaN). Record is marked as unusable.
- `sampler_degeneracy`: the degeneracy fields described in Section 4.5 below, computed against the original adjacency with the dropped-node mask. Mechanism-agnostic by design.
- `seed`: the deterministic sub-seed used for this replicate's sampler call.

Orchestrator output schema. The orchestrator (`build_degeneration_corpus`) flattens the per-replicate records into a single rectangular DataFrame. Each row corresponds to one (network, ρ, rate, replicate, mechanism) combination, with mechanism varying fastest in row order so that the two mechanism rows for the same cell are adjacent. The output columns are: `network_name`, `nominal_rho`, `nominal_rate`, `replicate_idx`, `mechanism` (`:full_removal` or `:outgoing_only`), `seed`, `dropped_nodes`, `realized_rate`, `realized_rho`, `bisection_status`, and the seven flattened degeneracy fields (`gc_fraction_of_remaining`, `n_observed`, `n_edges_observed`, `gc_collapse`, `too_small`, `no_edges`, `any_topo_degenerate`).

Three-source degeneracy flags. A degraded network can be degenerate for reasons the sampler can detect (topological collapse from dropped-node selection alone), for reasons the measurement code detects (non-finite or undefined measure values), or for reasons the framework detects (empty E/I bins, β-solver non-convergence). Each source flags its own degeneracies in its own record field: `sampler_degeneracy` (this section), `measure_degeneracy` (populated when the measure battery runs), and `framework_degeneracy` (populated when the framework runs). No code is forced to import another's internal state to detect degeneracies, and downstream summaries can filter on any combination of flags.

4.5 Degeneracy Detection

The sampler is responsible for detecting and recording topological degeneracies — those determinable from the dropped-node set alone, without invoking the measure code or the framework. Three quantities are computed, and the locked thresholds applied below.

Giant-component fraction (of remaining nodes). The largest connected component of the degraded network, expressed as a fraction of the *remaining* (not original) node set. The threshold is 0.30: below this, the degraded network has fragmented into pieces too small to support a global network analysis in the operational sense. The denominator is remaining-nodes rather than original-nodes because the question is whether what survives is structurally coherent, not how much of the original survives. Recorded as a continuous value alongside the threshold-derived Boolean.

Remaining node count. The number of nodes left after removal. The threshold is 25: below this, centralization standard deviations become too noisy and the framework's binning has too few nodes per bin for the procedure to be meaningful. The 50% rate cell on Moreno (N=70) leaves 35 nodes, comfortably above this floor; the threshold catches catastrophic cases (extreme high-rate cells on the smallest networks) without firing on the design grid.

Remaining edge count. The number of edges left after removal. The threshold is 1: a network with zero edges is degenerate by any reasonable definition. Possible at very high rates on sparse small networks (e.g., Toledo at 50% with strongly positive ρ).

Continuous values are the primary record. All three quantities are recorded as their raw continuous values, with the threshold-derived Booleans (`gc_collapse`, `too_small`, `no_edges`, and the disjunctive `any_topo_degenerate`) as convenience-summary views. This permits post-hoc re-thresholding without re-running the sampler: if the chosen thresholds turn out to be too strict or too lenient on the realized corpus, the cell-level degeneracy rates can be recomputed from the recorded continuous values rather than from a re-run of 48,000 draws.

Layer separation. The sampler does *not* attempt to detect measurement-time degeneracies (non-finite closeness on a fragmented network, zero-variance centralization, Bonacich non-convergence) or framework-internal degeneracies (empty E/I bins, β-solver failure). Those belong to the measure code and the framework respectively, recorded in their own degeneracy fields when each runs. The sampler's contract is narrow: it returns dropped nodes and topological flags. Composition of the three flag sources happens at analysis time, not in the sampler.

4.6 Sampler Validation Harness

Before running the sampler against the corpus, an eight-test harness verifies that it does what its API claims. The harness follows the convention of the Phase 0 test suites (e.g., `run_synthetic_path_centrality_tests`): a single driver function that runs all checks, prints a one-line status per check, and returns a Boolean for use in regression. Each check is callable in isolation for debugging.

Test 1: Bisection convergence on a tractable network. On Scotland (undirected, $N = 108$, moderate skew), target $\rho = +0.25$ at rate 0.10. Draw 100 replicates. Assert that the mean realized ρ across the 100 draws matches target within tolerance 0.04. The tolerance is loosened from a deterministic-bisection ideal because two layers of Monte Carlo noise compose: the M=50 inner-sample bisection estimate has standard error roughly 0.015, and the 100-replicate harness mean adds another roughly 0.010, giving a triangle-inequality bound of about 0.04. Single-replicate realized ρ is noisy; the test is on the mean across many seeds, not on any single draw.

Test 2: Exact rate hit per replicate. Across a small grid of (rate, ρ) pairs covering the design grid on Scotland and Moreno, every replicate's realized rate must equal round(target_rate $\cdot$ N) exactly. Asserted per-replicate, not on the mean. A bug where the ρ-targeting accidentally perturbed the rate would be subtle in mean terms but caught immediately by an exact-per-replicate check.

Test 3: Achievable-ρ ceiling detection. On a constructed 50-node star fixture (one hub with degree $N-1$, all leaves with degree 1 — the extreme centrality-concentration case), target $\rho = +0.95$ at rate 0.10. The bisection is expected to detect that the centrality distribution cannot support the target: the $b = 1$ probe estimate falls below 0.95, the saturation path activates (only one strictly positive prob entry against $k = 5$ required draws), and the sampler returns `bisection_status = :ceiling_hit`. The test asserts (a) 20/20 replicates report `:ceiling_hit`, and (b) the mean realized $|\rho|$ at the ceiling falls below the target. This is the most consequential test in the harness: a sampler that silently reports nominal targets it did not achieve poisons all downstream calibration analysis.

Test 4: Seed reproducibility. On Moreno (directed), ten replicate pairs are drawn with identical (N, target_rate, target_rho, s Each pair must produce bit-identical `dropped_nodes`, `realized_rate`, `realized_rho`, and `bisection_status`. This catches the entire class of sub-seed plumbing bugs: any non-determinism (whether from missing seed parameters, leaked RNG state, or thread-scheduling sensitivity in the Monte Carlo bisection's inner seeds) produces mismatches across the ten pairs.

Test 5: MCAR baseline. On Scotland, target $\rho = 0$ at rate 0.10. The Monte Carlo bisection's fast-path short-circuits to $b = 0$ (pure uniform sampling) without entering the bisection loop. The test asserts (a) all 100 replicates report `bisection_status = :converged`, and (b) the mean realized ρ across the 100 draws is near zero (within tolerance 0.04). This is the sanity check that $\rho = 0$ actually corresponds to MCAR in practice.

Test 6: Constant centrality fails explicitly. On a constructed 50-node 4-regular ring fixture (every node has degree 4 by construction, so centrality is exactly constant), target $\rho = +0.25$ at rate 0.10. The Monte Carlo bisection's inner correlation estimate returns NaN on the first iteration — a constant-against-anything correlation is undefined — and the bisection short-circuits to `:failed_other`. The test asserts (a) all 10 replicates report `:failed_other`, (b) all `realized_rho` values are NaN, and (c) all `dropped_nodes` vectors are empty. This replaces the prior design's "ties attenuate selectivity" assertion: the framework is loud about structural impossibility rather than producing a degraded-but-finite realized ρ on degenerate input.

Test 7: Degeneracy flagging fires on stress cells. On Moreno (directed, $N = 70$) at $50\% \times \rho = +0.50$ proved insufficiently adversarial to fire the topological-degeneracy flags during empirical testing: with 35 remaining nodes Moreno's clustering preserves enough connectivity to escape the `gc_collapse` and `too_small` thresholds. The empirically calibrated stress cell is Moreno at rate $= 0.65$ with target $\rho = +0.75$, which drops 46 nodes and leaves 24 below the `min_n` threshold of 25. The test draws 100 replicates and asserts that at least 50% trip `any_topo_degenerate`. The threshold-of-50% catches detection regressions while accommodating stochastic variation in which specific dropped sets actually fragment the remainder; in practice the cell fires on essentially every replicate (`too_small` on 100/100, `gc_collapse` on roughly 25/100). A silent failure here would produce clean grid summaries hiding a real failure mode.

Test 8: Outgoing-only materializer contract. On Moreno at $\rho = +0.25$, rate 0.10, one replicate is drawn. Both materializers (`apply_missingness` and `apply_missingness_outgoing_only`) are called on the same dropped-node set, and four assertions verify the SMM-style mechanism: (a) selection invariance — the dropped-node identifiers match across the two materializations; (b) roster preservation — the outgoing-only output's nodes table has the same row count as the input, with non-respondents retained as zero-out-degree actors; (c) src-only edge filtering — no edge in the outgoing-only output has `src` in the dropped-node set, while every original edge from a respondent to a dropped node is preserved; (d) undirected guard — calling `apply_missingness_outgoing_only` on Scotland (undirected) raises `ArgumentError`, since the SMM mechanism is undefined on undirected networks. This test exercises the doubled-mechanism architecture's structural correctness independent of any stochastic sampler behavior.

Small-network default, Marvel opt-in. The harness runs by default on Scotland, Moreno, the constructed star, and the constructed 4-regular ring — all small enough that the 100-replicate inner loops complete in a few seconds. Marvel is excluded from the default suite: sampler correctness is network-size-invariant (if it works on Scotland it works on Marvel, just slower), so the default suite preserves the property that it can actually be run every time. A separate diagnostic, `run_marvel_stress_test`, exercises the Marvel-scale behavior in detail and is described in the next subsection.

4.7 Marvel Stress Test

The harness above verifies sampler correctness on small networks where the per-test wall-clock cost is negligible. Marvel (N = 6,486, E = 77,227) raises questions the small-network harness cannot answer: per-replicate timing at corpus-large scale, bisection convergence on a centrality vector with realistic variance, degeneracy-detector behavior on a large clustered network, and the budget profile for a full validation grid. A separate diagnostic, `run_marvel_stress_test`, was run once after the harness passed to characterize this behavior.

Per-replicate timing. The centrality computation (one-time per network, cached for all replicates) completes in 0.029 seconds on Marvel, with mean centrality 23.8 matching the expected $2E/N = 23.81$ to four significant figures. Per-replicate wall-clock at the $(\rho = +0.25, \text{rate} = 0.10)$ design cell averages 0.147 seconds across 20 draws, with range 0.124–0.183 seconds. This is dominated by the Monte Carlo bisection's $\sim$1500 inner sampler operations at Marvel scale; the materialization and degeneracy check are sub-millisecond.

Bisection convergence on realistic centrality. All 20 replicates converged cleanly: zero `:ceiling_hit` and zero `:failed_other` statuses. The mean realized ρ was 0.233 against target 0.25, a deviation of 0.017 from target with per-replicate standard deviation 0.022 (about a fifth of Scotland's per-replicate variance, consistent with the larger N producing tighter sampling distributions). The standard error of the 20-replicate mean is 0.005, so the deviation is within the M=50-bisection-noise expectation; bisecting against an M=100 inner estimate would tighten this further if it ever proves necessary.

Degeneracy-detector engagement. A four-rate sweep (10%, 30%, 50%, 70%) at $\rho = +0.25$ with 10 replicates per cell showed the detector engages cleanly at high rates: 0% engagement at rates 0.10 and 0.30, 100% engagement at rates 0.50 and 0.70 with the `gc_collapse` flag firing. Crucially, `too_small` did *not* fire even at rate 0.70 (where 1,945 nodes survive, still well above the `min_n` threshold of 25): on Marvel-scale networks the detector's structural threshold (giant-component fraction) catches degeneracy that the node-count threshold misses. The saturation path was never triggered, confirming that Marvel's centrality distribution has more than enough variance to support the full k positive entries at $b = 1$.

Budget profile. At mean 0.147 seconds per replicate and a single-network grid of $5 \times 6 \times 100 = 3000$ cells, the per-network wall-clock estimate for Marvel is 7.4 minutes for one mechanism. Scaling to the sixteen-network corpus gives a single-mechanism corpus budget of approximately 2 hours and a doubled-mechanism corpus budget of approximately 4 hours, dominated by Marvel and Balikatan with the smaller networks contributing negligible time. These budgets fit comfortably within an overnight run on a single workstation.

4.8 Missingness Rates

The six missingness rates of Factor 1 (5%, 10%, 15%, 25%, 40%, 50%) are applied to every network. The same rate set and the same five induced-ρ levels apply to both weighted and unweighted networks; the framework's internal behavior differs between them (Stage 2 edge-weight reversal is skipped for unweighted networks) but the degeneration conditions are uniform across the corpus.

4.9 Reconstruction Procedure

Once nodes are selected for removal, they are removed from the network entirely — including all edges incident to them. This is the listwise deletion procedure standard in the network sampling literature (see Smith et al., 2022; Galaskiewicz, 1991; Costenbader and Valente, 2003; Borgatti et al., 2006). The resulting reduced network is G_{obs}, which is passed to the framework and the comparison baselines.

Note that this means the framework is not given any information about which specific nodes were removed (i.e., no list of partially observed nodes is provided in the basic protocol). This represents the most general case where the analyst knows missingness has occurred but does not know which specific nodes are missing. A supplementary set of conditions will also be run where the framework is given the list of nominated non-respondents (those who appear in observed edges but were removed), to evaluate the contribution of Stage 0.5 imputation.

Weighted versus unweighted networks. For weighted networks, both node-level (π_{node}) and edge-weight-level (π_{edge}) missingness are in principle modelable; the validation focuses on node-level missingness following the protocol above, with $\pi_{\text{edge}} = 0$ throughout to keep the design tractable. For unweighted networks, π_{edge} is structurally inapplicable as specified in the framework specification, and only π_{node} varies.

4.10 User-Supplied Prior

For each combination of network, induced ρ, and missingness rate, the framework is run with three different user-supplied ρ values to evaluate calibration sensitivity:

- **Correct prior:** the user-supplied ρ matches the induced ρ. This is the framework's best case and establishes its ceiling performance.
- **MCAR-assuming prior:** the user-supplied $\rho = 0$, regardless of the true missingness mechanism. This represents the case where the user has no domain knowledge or assumes the simplest case.
- **Opposite-sign prior:** the user-supplied ρ has the opposite sign from the induced value, with the same magnitude. This is the framework's worst case and tests how badly the framework fails when the user is fundamentally wrong about the missingness mechanism.

4.11 Iterations

For each cell of the (network, induced ρ, missingness rate, user-supplied ρ) grid, the missingness-induction-and-evaluation cycle is repeated 100 times for small and mid-sized networks, with a smaller count permitted for Marvel where per-replicate cost is highest and for cells whose degeneracy profile (Section 3.5 sampler design) makes additional replicates uninformative. The count is reduced from the 1,000 iterations used in Smith et al. (2022) to accommodate the framework's posterior sampling ($B = 1{,}000$ replicates per run). With 100 iterations per cell, coverage estimates have approximate standard error $\sqrt{0.95 \cdot 0.05/100} \approx 0.022$, which is acceptable for the validation's purposes. Because each replicate records its realized (rate, ρ) alongside the nominal target, coverage can be aggregated either against the nominal grid or against the realized conditions; the latter is the more honest summary on skewed networks where the achievable-ρ ceiling prevents the nominal target from being reached.

5 Comparison Baselines

The framework is compared against five baselines representing the realistic alternatives a practitioner would consider.

Listwise deletion (floor baseline). The reduced network G_{obs} is used directly without imputation. Measures are computed on G_{obs} as-is. This represents the no-method case and establishes the magnitude of the missing-data problem the framework is addressing.

Asymmetric imputation (directed networks only). Following Smith et al. (2022): when an observed forward edge points to a removed node, no reverse edge is added. This privileges observed structure at the cost of systematic asymmetry.

Symmetric imputation. Following Smith et al. (2022): when an observed forward edge points to a removed node, a reverse edge is assumed to exist. For directed networks, this assumes all ties are reciprocated. For undirected networks, this is the natural reconstruction.

Probabilistic imputation (directed networks only). Following Smith et al. (2022): reverse edges to removed nodes are added with probability equal to the global reciprocity rate computed from respondent-respondent dyads in G_{obs}. Repeated 100 times per missingness realization, with point estimates summarized as means across the 100 imputations.

Random augmentation bootstrap (Chapter 4 baseline). Following Bellutta (2026)'s Chapter 4: edge weights are added uniformly at random to G_{obs} until the total weight reaches an estimated true level. Repeated $B = 1{,}000$ times with credible intervals computed from the resulting distribution. This is the simplest bootstrap baseline and represents the prior bootstrap approach the framework is designed to improve upon.

Model-based imputation methods (simple and complex ERGM-based) are deliberately excluded from the comparison. Smith et al. (2022) established their performance characteristics on the same general class of networks; their computational cost scales poorly to networks the size of Balikatan; and the framework's positioning is as a scalable alternative to model-based imputation rather than as a competitor to it.

6 Measures and Evaluation Criteria

The validation evaluates each method on the same network measures used in Smith et al. (2022), allowing direct comparability with that prior work. Evaluation is conducted on two tracks: point-estimate accuracy (comparable to prior imputation studies) and credible-interval calibration (the framework's distinctive contribution).

6.1 Measures

The Phase 1 measure set is locked to the battery below. Each measure is computed in a weight-aware manner: for weighted networks the degree family uses strength (weight-summed degree) and the path-based measures (closeness, betweenness, mean inverse distance) use tie-strength edge semantics (Dijkstra on $1/w$, so stronger ties are shorter); for unweighted networks the degree family is the binary degree and the path-based measures use the binary (`:ignore`) semantics that reproduce Smith et al. (2022)'s convention. Bonacich power is already weight-aware and is computed identically in both cases. Global transitivity is computed by a direct $O(\sum_v d_v^2)$ triangle-counting algorithm in Phase 1 rather than by the triad census used for the Phase 0 characterization tables; the two agree on small networks but the direct algorithm scales to Marvel.

Node-level measures (centrality).

- In-degree / in-strength centrality (directed networks only)
- Out-degree / out-strength centrality (directed networks only)
- Symmetric degree / total strength centrality
- Closeness centrality (tie-strength weighted; binary for unweighted)
- Betweenness centrality (tie-strength weighted; binary for unweighted)
- Bonacich power centrality

Network-level measures (centralization).

- In-degree / in-strength centralization (directed networks only)
- Out-degree / out-strength centralization (directed networks only)
- Symmetric degree / total strength centralization
- Closeness centralization
- Betweenness centralization
- Bonacich power centralization

Topology measures.

- Largest component size (connected / weakly connected component)
- Largest bicomponent size (biconnected component)
- Mean inverse geodesic distance (tie-strength weighted; binary for unweighted)
- Global transitivity (direct triangle-counting algorithm)

Measures excluded from Phase 1. Two measures used in the Phase 0 characterization are deliberately *not* carried into the Phase 1 validation grid. The ranked-cluster tau statistic (Tau_{RC}) is a Phase 0 descriptive only: it requires a Monte Carlo null model per evaluation, which is prohibitively expensive at the scale of the validation grid, and it is undefined for undirected networks. Structural-equivalence blockmodeling is likewise excluded: its per-replicate cost does not reduce to the scalar-per-network form the validation grid requires. Both remain available as corpus-characterization tools but are out of scope for the degraded-network evaluation.

6.2 Track A: Point-Estimate Accuracy

For direct comparability with Smith et al. (2022), point estimates are evaluated using their established metrics:

Node-level measures. For each method and each missingness condition, compute the correlation between the true node-level centrality scores (computed on T) and the observed/imputed scores (computed on G_{obs} after each method's reconstruction). High correlation indicates the method preserves the relative ranking of nodes. For the framework, the point estimate is taken as the posterior median across the B replicates.

Network-level measures. For each method and each condition, compute the standardized absolute bias:

$$\text{Bias} = \left| \frac{m(G_{\text{obs}}^{\text{reconstructed}}) - m(T)}{m(T)} \right|.$$

For the framework, the point estimate is again the posterior median.

6.3 Track B: Credible-Interval Calibration

The framework's distinctive output is a posterior sample, summarized as a credible interval. Two evaluation metrics apply only to the framework (and to random augmentation, which also produces intervals):

Coverage. For each condition, the proportion of iterations in which the 95% credible interval contains the true measure value computed on T. Nominal coverage is 0.95. Coverage substantially below 0.95 indicates the framework's intervals are too narrow (overconfident); coverage near 1.00 indicates the intervals are too wide to be informative.

Width. The mean width of the 95% credible interval across iterations, normalized by an appropriate scale (the standard deviation of the true measure across nodes for node-level measures; the absolute value of the true measure for network-level measures). Narrower intervals are more useful provided coverage is adequate.

6.4 Track C: Calibration Sensitivity

The framework's calibration is also evaluated as the user-supplied ρ deviates from the induced ρ. Specifically, coverage and width are computed for each of the three prior conditions (correct, MCAR-assuming, opposite-sign) and compared. Graceful degradation — where coverage drops modestly and width increases moderately as the prior departs from truth — is a desired property. Catastrophic failure — where coverage drops to near zero under the wrong prior — would be a significant problem.

7 Success Conditions

The framework will be considered validated and ready for publication if the following conditions are met across the validation corpus.

7.1 Primary Conditions (must be met)

C1: Coverage under correct prior. Across all networks, induced ρ values, and missingness rates up to 30%, the framework's 95% credible intervals achieve at least 90% empirical coverage when the user supplies the correct ρ. (The 90% threshold rather than 95% acknowledges that the framework is an approximate Bayesian procedure, not a strict one.)

C2: Point estimates competitive with simple imputation. For directed networks at moderate missingness (10–25%), the framework's posterior median achieves bias and node-centrality correlation comparable to (within 10% of) the best-performing simple imputation method (probabilistic or symmetric, as appropriate) on each measure.

C3: Improvement over listwise deletion. Across all conditions, the framework's coverage and width are strictly better than listwise deletion combined with naive bootstrap (i.e., the random augmentation baseline) when the missingness mechanism is non-MCAR (induced $|\rho| \geq 0.25$).

C4: Scaling to Balikatan. The framework completes a single validation iteration on Balikatan ($N = 1{,}347$) in under 10 minutes of wall-clock time on a single modern CPU core, making the full validation grid computationally feasible.

C5: Unweighted network support. The framework operates correctly on all four unweighted classes (unweighted directed, unweighted undirected) across the corpus's unweighted networks. "Operates correctly" means: (a) the framework runs to completion without errors on these networks; (b) Stage 2 is correctly skipped (no edge-weight reversal attempted); (c) credible interval coverage under correct prior meets the C1 threshold (90%) on these networks. This condition formalizes the framework's distinctive ability to handle unweighted networks, which the Chapter 4 framework could not.

7.2 Secondary Conditions (desired but not strictly required)

C6: Coverage under MCAR-assuming prior. When the user supplies $\rho = 0$ but the true induced ρ is ± 0.25, coverage drops by no more than 10 percentage points relative to the correct-prior case. This indicates the framework is robust to mild prior misspecification.

C7: Graceful degradation under opposite-sign prior. When the user supplies a prior with the wrong sign, coverage may drop substantially but does not fall below 50% for any network and missingness rate up to 30%. This indicates the framework fails gracefully rather than catastrophically.

C8: E/I binning improves performance on betweenness. For networks with non-trivial community structure (modularity > 0.30), the framework with broker detection enabled achieves better coverage on betweenness centrality measures than a hypothetical degree-only variant.

7.3 Failure Conditions (would prompt revision)

F1: Coverage failure under correct prior. If C1 is not met for more than two networks in the corpus, the framework's coverage mechanism (logistic bin-assignment, Laplace smoothing, or sampling distributions) requires revision before publication.

F2: Catastrophic failure under wrong prior. If C7 is violated — coverage falls below 50% with an opposite-sign prior at moderate missingness — the framework's design is too sensitive to the prior and may not be suitable for general use.

F3: Failure to scale. If C4 is not met, the framework requires algorithmic optimization before validation can proceed at scale.

8 Validation Pipeline Architecture

The validation pipeline will be implemented in Python alongside the framework itself, following the module structure proposed in the accompanying implementation discussions. Key architectural decisions:

Data format. All networks are stored in a unified Parquet representation: a `nodes` table with node IDs and attributes, and an `edges` table with source, target, weight, and (for multiplex sources) layer identifier. This format is efficient for I/O, supports both Python and R/Julia analysis, and accommodates the eventual extension to multidimensional networks.

Reproducibility. All Monte Carlo iterations use seeded random number generators, with the seed determined deterministically from the condition specification (network, induced ρ, missingness rate, iteration index). This ensures the validation can be reproduced exactly and that individual conditions can be re-run without re-running the full grid.

Output. Framework runs produce per-iteration outputs in Parquet format: posterior samples for each measure, point estimates, credible interval bounds, and diagnostic information (realized correlation, binning mode used, etc.). Comparison method outputs are stored in parallel structures. Analysis and figure generation are performed in R using the `arrow` package to read the Parquet output.

Parallelization. The validation grid is embarrassingly parallel across (network, induced ρ, missingness rate, replicate) combinations. The Julia orchestrator (`build_degeneration_corpus`) uses `Threads.@threads :static` with a sampler-record cache that ensures both materialization mechanisms on the same cell share the underlying sampler call. The degeneration grid comprises approximately 72,000 records (the single-mechanism $16 \times 5 \times 6 \times 100 = 48{,}000$ records plus the second mechanism on directed networks only). At the Marvel-calibrated mean per-replicate cost of 0.147 seconds, the full doubled-mechanism degeneration grid completes in approximately 4 hours on a single workstation with sufficient threading. The downstream framework-evaluation grid — where each degeneration cell is multiplied by user-supplied ρ variations and B posterior replicates per measure — is correspondingly larger and is budgeted separately at the Track-A/B/C planning stage.

9 Reporting Plan

The final validation report will be a separate document, structured around the success conditions:

1. Corpus characterization (Phase 0 results, descriptive statistics for all sixteen networks)
2. Performance under correct prior (C1, C3): coverage, width, bias, and node-correlation results across the full grid
3. Comparison with simple imputation (C2): head-to-head point-estimate comparison
4. Calibration sensitivity (C6, C7): performance under MCAR-assuming and opposite-sign priors

5. Special analyses (C5, C8): unweighted network support and the value of E/I binning; the value of Stage 0.5 imputation when nominated non-respondents are identified
6. Scaling and computational performance (C4)
7. Discussion and limitations

Figures will follow the visual conventions of Smith et al. (2022): small-multiples decision-tree style figures showing the best method by condition, supplemented by coverage and width plots across the missingness-rate axis.

10 Risks and Contingencies

Risk 1: Framework coverage is too low under correct prior. If C1 fails systematically, the most likely causes are (a) the bin-assignment distribution does not produce the intended correlation in practice, (b) the rank-rank matrix is too smoothed (overly Laplace-regularized), or (c) the Poisson weight model is wrong for some networks. Diagnostic outputs will help identify which is the issue. Mitigations include refining the β-solving procedure, reducing smoothing, or adopting weight distributions tuned to the input network.

Risk 2: Computational cost exceeds the budget. If the full validation grid takes longer than anticipated, the first cut is to reduce iterations per cell from 100 to 50 (raises the standard error on coverage estimates from ≈ 0.022 to ≈ 0.031 but remains usable for the primary conditions). The second cut is to reduce missingness rates from six to four (drop 5% and 50%, keeping 10%, 15%, 25%, 40%). The third cut is to reduce posterior samples from $B = 1{,}000$ to $B = 500$. The per-cell degeneracy profile (Section 3.5 sampler design) provides a fourth, targeted lever: cells that degenerate at high frequency need few replicates, so replication can be trimmed cell-by-cell rather than uniformly.

Risk 3: Balikatan's structure is anomalous. If Phase 0 reveals that Balikatan has structural properties that make it an outlier in the corpus (e.g., extremely heavy-tailed degree distribution, multiple disconnected components), the validation may need to treat Balikatan as a case study rather than including it in the main factorial. The reporting structure accommodates this contingency.

Risk 4: Empirical centralization values cluster too tightly. If Phase 0 reveals that all five empirical networks have similar centralization, the synthetic networks become the primary source of centralization variation. The synthetic design has flexibility built in to address this if needed.

11 Milestones

The validation is organized into seven milestones. Each milestone is expressed as one or more *necessary conditions for completion* — empirical or technical facts that must hold before subsequent work can responsibly proceed. The conditions are deliberately falsifiable: if the supporting evidence does not establish the condition, the milestone is not met, and the work either iterates or halts pending revision.

Milestone 1: Data Infrastructure

Necessary conditions.

- All five empirical networks load into the unified Parquet format without loss of information.

- Loaded networks reproduce published descriptive statistics (node count, edge count, density, direction) within rounding tolerance.

- The unified format accommodates all sources (five empirical, plus synthetic networks generated in Milestone 3) without source-specific schema variations.

Artifact. `nodes.parquet` and `edges.parquet` files for each network, plus a loader-validation report comparing loaded statistics against source documentation.

Gates. No subsequent milestone can proceed until the data is verified loaded; corruption or schema mismatch discovered later contaminates everything downstream.

Milestone 2: Empirical Corpus Characterization (Phase 0)

Status: complete. The eight Phase 0 tables (Tables 2–9) report the full characterization, and the conditions below are met.

Necessary conditions.

- Descriptive statistics from Section 3.5 (centralization, transitivity, mean degree, weight distribution, triad census, etc.) are computed for all five empirical networks and reported in the Phase 0 tables.

- The empirical corpus collectively spans enough structural variation to make the validation meaningful — in particular, at least two distinct centralization regimes must be represented across the empirical networks. (Met: degree centralization ranges from the near-uniform small networks to the strongly hub-skewed Balikatan and Marvel cases.)

- Any empirical network that turns out to be structurally anomalous (e.g., disconnected, trivially small communities, degenerate degree distribution) is identified and a remediation decision is made (include with caveats, treat as case study, or exclude). (Met: Marvel's largest component spans 70% of nodes; it is included with this fragmentation noted.)

Artifact. Phase 0 report: the six descriptive tables in Section 3.5, an empirical analog of Smith et al. (2022)'s Table 1 extended with weighted and triad-census characterizations, plus the narrative interpretation of structural coverage in that section.

Gates. The Phase 0 report determined that the synthetic network designs did not require redesign; the empirical corpus, with Marvel added at the large-clustered extreme, spans the intended structural range and the synthetic generators anchor the controlled centralization endpoints as designed.

Milestone 3: Synthetic Network Generation and Unweighted Variants

Status: complete. All synthetic networks and unweighted threshold variants are generated and present in the corpus (Table 1).

Necessary conditions.

- Synthetic 1 and Synthetic 2 are generated in both weighted and unweighted variants, each in directed and undirected forms (eight synthetic networks total).

- Thresholded unweighted variants of Moreno, Toledo, and Marvel are produced from the loaded weighted versions.
- Each synthetic network's actual structural properties (computed post-generation) fall within the target ranges specified in Section 3.2 or are explicitly noted as deviations with justification.
- Together, the empirical and synthetic networks span low, moderate, and high centralization, both small ($N \leq 250$) and larger ($N \geq 600$) sizes, and all four cells of the weighted $\times$ directed design with at least three networks per cell.

Artifact. Synthetic and unweighted-variant network files in the unified Parquet format, plus a generation report comparing target and actual structural properties.

Gates. The full corpus must be finalized before the framework implementation enters validation testing, so that testing covers the network suite that will be used in production validation.

Milestone 4: Framework Implementation

Necessary conditions.

- All algorithmic components specified in the framework specification are implemented: bin assignment, P and R matrix computation, Louvain community detection with E/I scoring, β-solving for ρ-conditional bin distributions, Stage 0.5 imputation, Stage 1 node augmentation, Stage 2 edge-weight reversal, posterior aggregation.
- Unit tests cover each component with known-output cases, including edge cases (unweighted networks, $\pi_{\text{node}} = 0$, $\pi_{\text{edge}} = 0$, single-community fallback, sparse cell handling).
- End-to-end smoke test on a small network (Moreno) produces output that is internally consistent: realized correlations between bin assignment and missingness across replicates match the requested ρ within tolerance; bin counts match N_{add} in expectation.
- Computational performance permits at least one full framework run on Balikatan ($N = 1{,}347$) in under 10 minutes wall-clock on a single modern CPU core (condition C4 from Section 6.1).

Artifact. Implemented framework as a Python package, unit test suite, and a smoke-test report demonstrating end-to-end operation and the Balikatan timing benchmark.

Gates. Validation cannot run on an unimplemented or unverified framework. The Balikatan timing constraint must be met before committing to the full validation grid; if not met, optimization work intervenes before Milestone 5.

Milestone 5: Comparison Baseline Implementation

Necessary conditions.

- Listwise deletion, asymmetric, symmetric, and probabilistic imputation methods are implemented and produce results matching Smith et al. (2022)'s reported behavior on Moreno when supplied with equivalent missingness conditions.

- Random augmentation bootstrap (from Bellutta (2026), Chapter 4) is implemented and produces results matching the Chapter 4 description.
- All baseline methods accept the same network and missingness inputs as the framework, producing comparable outputs (point estimates for all; credible intervals for random augmentation and the framework).

Artifact. Implemented baseline methods as part of the validation pipeline, plus a baseline-verification report showing that each baseline reproduces published behavior on at least one network.

Gates. The validation cannot make valid comparisons against baselines that have not been verified to behave correctly. Verification against published results is the necessary check.

Milestone 6: Pilot Validation

Necessary conditions.

- The full validation pipeline (network loading, missingness induction, framework execution, baseline execution, metric computation, output aggregation) runs end-to-end on Moreno across all five induced ρ values, all six missingness rates, and all three user-supplied ρ conditions, with reduced iterations ($n = 50$) for tractability.
- Pilot results are sane: coverage is in a plausible range (not 0 or 1 everywhere), interval widths scale reasonably with missingness, and method comparisons produce expected ordering on at least one condition (e.g., listwise deletion bias increases with missingness rate).
- No catastrophic failures are observed: no crashes, no NaN outputs, no negative interval widths, no coverage outside $[0, 1]$.

Artifact. Pilot validation report on Moreno: coverage and width tables, point-estimate comparisons, and an explicit assessment of whether the pipeline is producing sensible results.

Gates. The pilot exists specifically to catch problems before committing computational resources to the full grid. If pilot results are not sane, the full validation does not proceed; the issue is diagnosed and the pipeline revised.

Milestone 7: Full Validation and Reporting

Necessary conditions.

- The full factorial validation grid completes across all sixteen networks, five induced ρ values, six missingness rates, three user-supplied ρ conditions, and up to 100 iterations per cell (fewer where the degeneracy profile or Marvel's per-replicate cost warrants).
- All measures from Section 5.1 are computed for the framework, all applicable baselines, and the true network values.
- Output is stored in the agreed Parquet schema with sufficient metadata to reconstruct any individual condition's run.
- The primary success conditions (C1–C5 from Section 6.1) are evaluated against the empirical results, with each condition's status (met / partially met / not met) documented.

- The secondary conditions (C6–C8) and failure conditions (F1–F3) are evaluated and documented.
- A validation report is drafted following the reporting plan in Section 8, with figures generated from the Parquet output.

Artifact. Complete validation dataset in Parquet format, plus a validation report addressing every success condition and providing the comparison narrative against the baselines.

Gates. The framework is considered validated and ready for broader use only if the primary conditions (C1–C5) are met. If any primary condition fails, the failure analysis in Section 9 determines next steps (revision of framework, revision of validation, or limitation acknowledgment in publication).

Summary of Milestones and Dependencies

#	Milestone	Primary artifact	Gates the start of
1	Data infrastructure	Loaded Parquet files	M2, M3, M4
2	Empirical characterization (Phase 0)	Phase 0 report	M3
3	Synthetic network generation	Synthetic Parquet files	M6
4	Framework implementation	Python package + smoke test	M5, M6
5	Comparison baseline implementation	Verified baseline code	M6
6	Pilot validation	Pilot report on Moreno	M7
7	Full validation and reporting	Validation report	Publication / release

Table 10: Milestone dependency structure. Each milestone's completion gates the start of one or more later milestones.

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